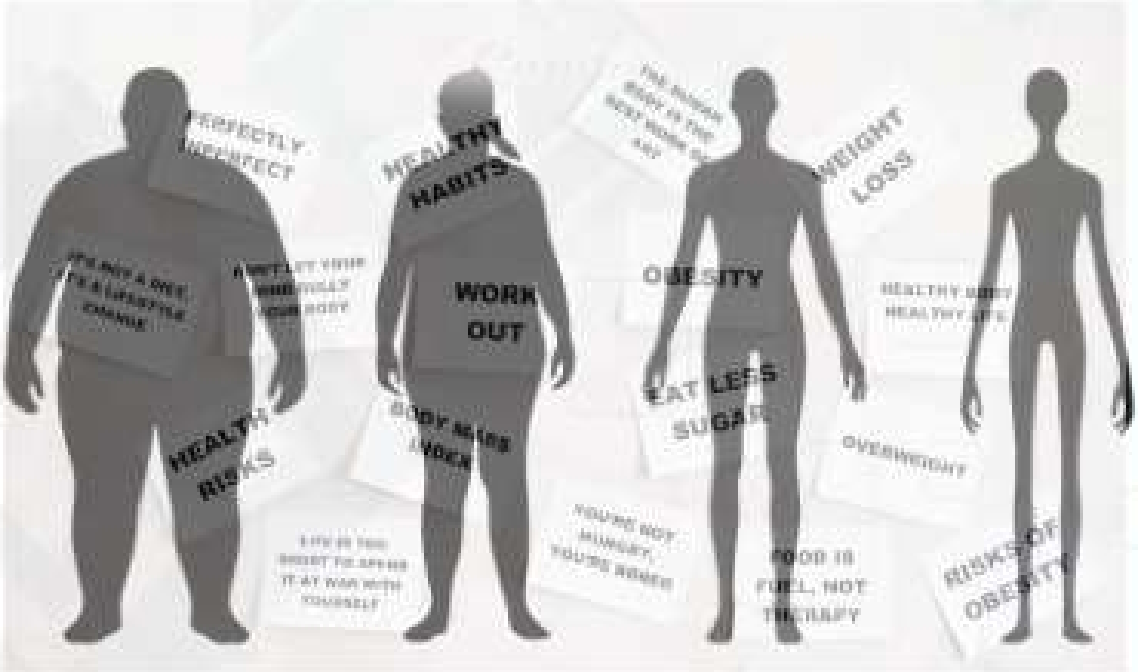
VIT BHOPAL

BMI-CALCULATOR

VITYARTHI PROJECT

FUNDAMENTALS OF AI&ML



SUBMITTED BY: MANYA GUPTA

REGISTRATION NO:: 25BAI10407

INTRODUCTION & PROBLEM STATEMENT

The Body Mass Index (BMI) Calculator is a web-based application designed to evaluate a person's body weight relative to their height. In today's fast-paced lifestyle, individuals often neglect regular health monitoring. Many people are unaware of whether their weight falls within a healthy range, which can lead to serious health issues such as obesity, malnutrition, or lifestyle diseases.

The problem addressed by this project is the lack of a quick, accessible, and easy-to-use tool for calculating BMI. While manual calculation is possible, it is inconvenient and prone to error. Therefore, this project aims to provide a simple digital solution that allows users to input their height and weight and instantly receive their BMI along with an interpretation of their health status.

This project demonstrates how basic web technologies can be used to solve real-world problems effectively and efficiently.

Functional Requirements

The BMI Calculator system must fulfill the following functional requirements:

- Accept user input for height in centimeters
- Accept user input for weight in kilograms
- Validate that both input fields are not empty
- Convert height from centimeters to meters
- Perform BMI calculation using the formula
- Display the calculated BMI value
- Categorize the BMI into standard categories (Underweight, Normal, Overweight, Obese)
- Display the corresponding category to the user
- Handle invalid inputs gracefully by showing appropriate error messages.
- These functionalities ensure that the system performs accurate calculations and provides meaningful output to the user.

Non-Functional Requirements

The non-functional requirements define how the system performs:

- The application must have a simple and user-friendly interface
- The system should respond quickly and display results instantly
- The design should be visually appealing and easy to understand
- The application should work across modern web browsers such as Chrome, Edge, and Firefox
- The code should be clean, modular, and easy to maintain
- The system should not require internet connectivity after loading (client-side execution)
- These requirements ensure usability, performance, and maintainability of the system.

System Architecture

The BMI Calculator follows a client-side architecture. It is a single-page web application where all processing is done on the user's device.

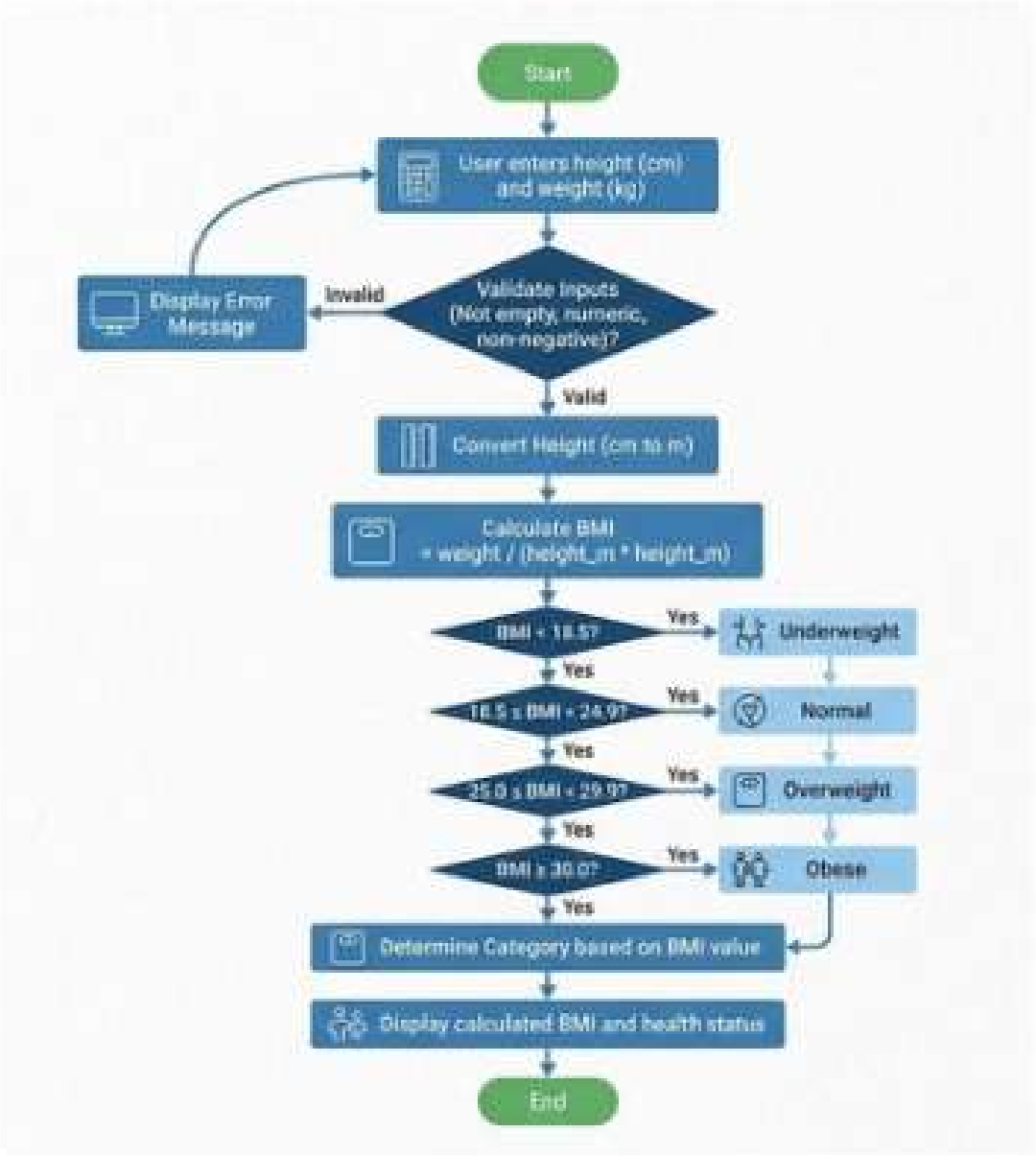
The system consists of three main components:

HTML (Structure) - Defines the layout of the webpage, including input fields, buttons, and display area.
CSS (Presentation) - Handles styling, layout, and visual appearance of the application.
JavaScript (Logic) - Performs calculations, input validation, and updates the output dynamically.

When the user enters values and clicks the calculate button, JavaScript retrieves the input, processes it, and updates the result on the webpage without reloading.

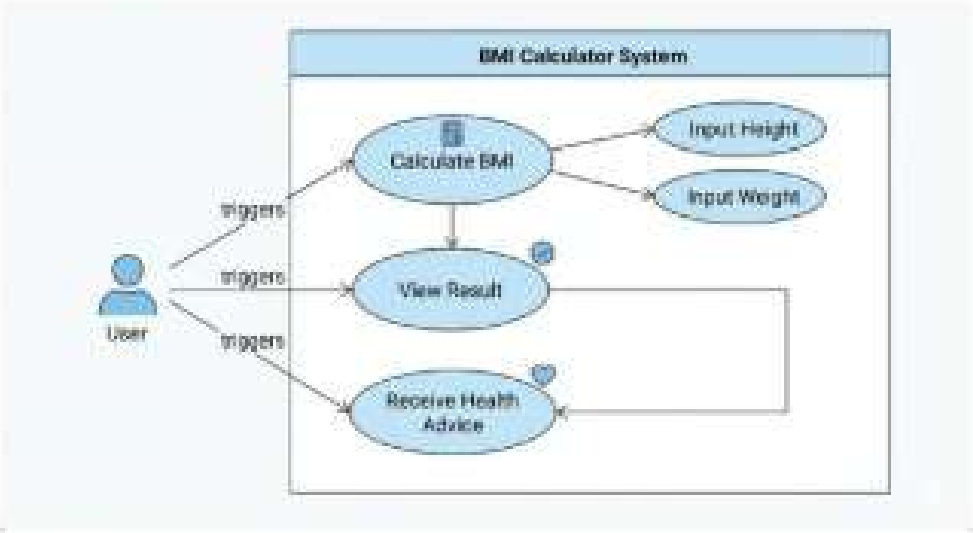
Design Diagrams

WORKFLOW DIAGRAM:

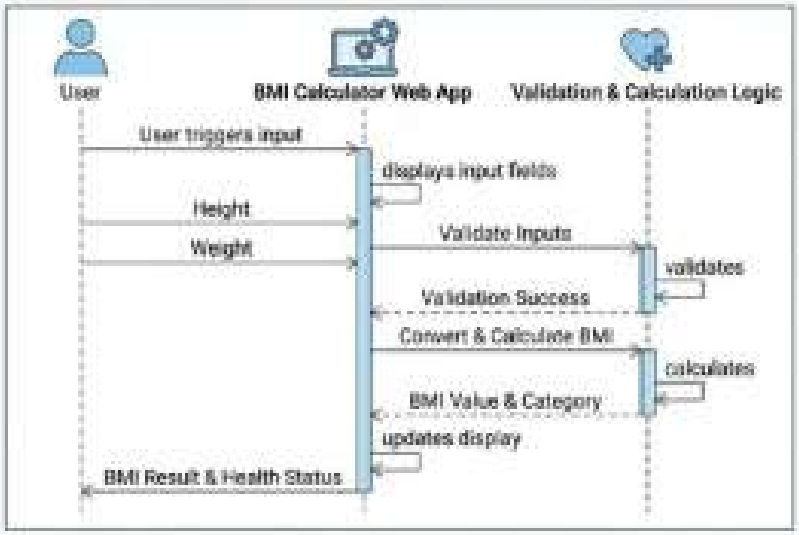


Design Diagrams

USE-CASE DIAGRAM:



SEQUENCE DIAGRAM:



Design Decisions & Rationale

Several design decisions were made to keep the project simple and efficient:

Client-side implementation was chosen to avoid complexity of backend systems
Vanilla JavaScript was used instead of frameworks to keep the project lightweight
Input validation was added to improve user experience and prevent errors
Minimal UI design ensures clarity and ease of use
Instant calculation improves responsiveness

These decisions help balance simplicity, performance, and usability.

Implementation Details

The implementation is divided into three parts:

HTML: Contains input fields for height and weight, a button to trigger calculation, and a section to display results

CSS: Provides styling such as layout alignment, colors, and spacing

JavaScript: Implements the logic

Retrieves input values using DOM methods

Converts height into meters

Calculates BMI

Uses conditional statements to determine category

Displays the result dynamically

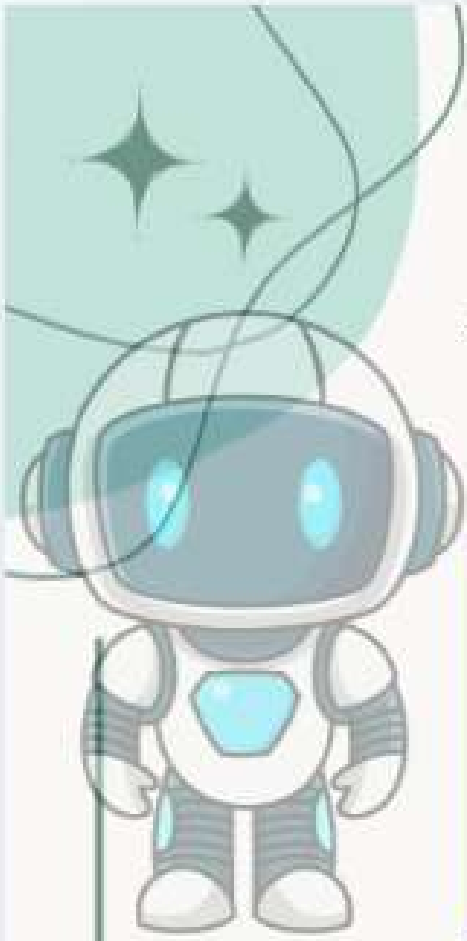
The application runs entirely in the browser without requiring any external libraries.

Testing Approach

The application was tested using different input values to ensure correctness:

Valid inputs were tested to verify accurate BMI calculation
Boundary values (e.g., BMI = 18.5, 24.9) were checked
Empty input fields were tested to confirm error handling
Different browsers were used to test compatibility

This testing ensures the reliability and accuracy of the application.



Screenshots & Results

Screenshots & Results

CONCLUSION

Challenges Faced

During development, several challenges were encountered:

- Handling empty input fields correctly
- Understanding DOM manipulation in JavaScript
- Debugging errors related to incorrect syntax
- Ensuring correct unit conversion
-

These challenges were resolved through testing and debugging.

Learnings & Key Takeaways

This project helped in understanding:

-
- Basics of web development (HTML, CSS, JavaScript)
- DOM manipulation and event handling
- Input validation techniques
- Logical problem solving using conditions
- Debugging and error handling

It also improved practical implementation skills.

Future Enhancements

The project can be improved by:

-
- Adding graphical representation of BMI
- Providing personalized health recommendations
- Making the UI more interactive with animations
- Adding mobile responsiveness
- Integrating with a database for user tracking

REFERENCES

- MDN Web Docs (HTML, CSS, JavaScript)
- W3Schools tutorials
- BMI formula reference from standard health guidelines